
A Technical Companion to “My ‘Next Generation’ Automated Theorem Prover”

Hilbert-Ordered Proof Search, Verified Inference,
Search Speed, and Restricted-Advice Guidance

Part One: <https://tinyurl.com/yscu3y3>

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Version 1.0 – Public Technical Edition – August 8, 2026

Scope and claim status. Classical Hilbert facts are used as background. The paper proves soundness and exhaustive positive-enumeration results for a stated finite-action model. It also proves two limited speed facts: uninformed no-repeat orderings have the same expected target rank under a uniform target, while a full-grid Hilbert path is transition-cost optimal under a declared one-coordinate incremental cost model. No universal time-to-proof optimality is claimed. The prior-art review includes Rostami–Bright (2025), which uses Hilbert literal ordering in proof-producing SAT; accordingly, the originality claim is restricted to the rule-specific inference-address construction developed here.

Abstract

This paper formalizes a central Data-ATP proposal: a Hilbert-style probe can be interpreted not merely as a ranking device, but as an exhaustive generator of candidate inference actions. The proof-theoretic vocabulary is taken from Tenneson’s *Depths of Induction and Formalized Self-Awareness* [13]. A k -ary inference rule is a partial function on ordered k -tuples of well-formed formulas. A direct consequence is the value of a defined rule instance, and a proof is a finite sequence of initial hypotheses and direct consequences of earlier lines. For a finite verified set C , each k -ary rule therefore induces a premise-choice space C^k , and heterogeneous rule spaces form a tagged coproduct. Finite Hilbert traversals exhaust power-of-two discretizations of these components and move between face-adjacent cells, so consecutive addresses differ in one coordinate. Decoding a visited address produces a candidate rule instance; if the rule is defined and verification accepts the result, the resulting formula is a legitimate next proof line.

The main results separate four claims that are often blurred. First, theorem-point soundness: every accepted generated formula is a direct consequence of already verified formulas and therefore lies in the consequence set. Second, complete-epoch correctness: a completed frozen epoch computes exactly the one-step direct-consequence closure of the frozen state, under explicit finite-action and verifier assumptions. Third, exhaustive positive completeness: repeated fair complete epochs enumerate every theorem having a finite derivation in the modeled calculus. Fourth, a finite-budget Data-ATP run has a two-outcome contract: it returns a verified proof certificate or reports budget exhaustion; budget exhaustion is not a proof of unprovability. The classical surjectivity of a Hilbert space-filling curve supplies the conceptual exact-coverage analogy, while the executable completeness proof uses finite discrete coverage rather than an impossible literal traversal of an infinite limiting curve [1, 2, 3].

The paper then separates several meanings of search speed. If a target is uniformly distributed over a finite feasible set, every exhaustive no-repeat ordering has the same expected target rank; Hilbert order cannot create information. Independent random sampling with replacement is strictly worse for guaranteed coverage because it revisits candidates. In a different cost model, however, finite Hilbert traversal has a provable optimality property: on a full power-of-two rule grid, every consecutive Hilbert cell changes exactly one coordinate by one unit, so the total ℓ_1 transition length and the total number of changed premise slots attain the minimum possible value among exhaustive orderings. This optimality is not unique, and it becomes irrelevant when adjacent candidates share no reusable computation.

The paper’s single conjecture concerns a restricted-advice “Professor.” The Professor may privately possess a valid proof while communicating only through a predeclared bounded channel, such as a few-bit progress score. The verifier still determines theoremhood. The conjecture asks whether such bounded privileged guidance can substantially reduce total cost-to-proof, with Professor computation included, without directly transmitting proof steps and without sacrificing exhaustive fallback coverage. This is related to the established study of algorithms with advice, but the application here is proof-search allocation rather than online competitive analysis [9]. The prior-art review found a close but materially different precedent in Hilbert-ordered Boolean literals for proof-producing SAT solving [8]. Accordingly, the originality claim is deliberately narrow. No matching construction was located in the reviewed literature in which Hilbert traversal itself exhausts tagged rule-specific inference-address spaces and is used to derive theorem-generation and closure results.

Keywords. automated theorem proving; proof search; Hilbert curve; space-filling curve; inference rules; proof certificates; machine-guided reasoning.

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1 From the public post to a research specification

The public essay and the preceding Hilbert-search manuscript supply the narrative and constructional starting points for this technical paper [12, 14]. The public essay begins with a deliberately simple observation: logic does not allocate attention. A formal system can say which moves are legal; a verifier can say whether a finished proof certificate is correct; neither statement alone tells a prover where to spend its next expansion. The Data-ATP research program treats that gap as an engineering and scientific object in its own right [12, 5, 6].

The public version develops five intuitions that will be retained exactly, but made more precise here:

1. candidate proof actions can be given addresses;
2. a rule’s premise arity gives a natural base dimension for its premise-choice space;
3. different rules belong in tagged components rather than in one meaningless Cartesian product;
4. a locality-preserving traversal can schedule addresses without owning mathematical truth; and
5. usefulness is an empirical question under fixed budgets, not an aesthetic consequence of the Hilbert curve.

The technical companion goes further in four directions. First, it states the finite-action construction and its coverage guarantees formally. Second, it separates address, feasibility, representation, geometry, and scheduling. Third, it incorporates the constrained hidden-target and creativity analyses developed in companion research manuscripts. Fourth, it performs a prior-art boundary review so that “original research” can be stated narrowly enough to be credible.

The question this paper actually asks

Can a structured representation plus a locality-preserving scheduler cause an ATP to find useful verified mathematics earlier, or at lower cost, than matched simpler schedules under the same formal information and computational budget?

2 Minimal formal-system background

The technical argument needs only a small part of the framework developed in *Depths of Induction and Formalized Self-Awareness* [13]. It is restated here so that the Hilbert construction can be read independently.

Definition 2.1 (Inference rule). Let y be a set of well-formed formulas. A k -ary inference rule on y is a partial function

$$r : y^k \rightharpoonup y,$$

where $k \geq 1$. Its arity is $\text{ar}(r) = k$. The partiality is essential: some ordered premise tuples license the rule and others do not.

Definition 2.2 (Formal system). A formal system is an ordered triple

$$F = (x, y, z).$$

Here x is a nonempty alphabet. The set y consists of finite strings over x designated as well-formed formulas, and z is a set of finitary inference rules on y .

Definition 2.3 (Direct consequence). Let $A \subseteq y$. A formula $\beta \in y$ is a direct consequence of A if there exist a rule $r \in z$ of arity k and a tuple $(\alpha_1, \dots, \alpha_k) \in A^k$ such that

$$r(\alpha_1, \dots, \alpha_k) = \beta.$$

Definition 2.4 (Proof and consequence set). Let $\Gamma \subseteq y$. A proof from Γ is a finite sequence

$$\pi = \langle \pi(1), \dots, \pi(n) \rangle$$

such that each line $\pi(j)$ either belongs to Γ or is a direct consequence of earlier lines. Write $\Gamma \vdash_F \varphi$ when a proof from Γ ends in φ , and define

$$\text{Con}_F(\Gamma) = \{\varphi \in y : \Gamma \vdash_F \varphi\}.$$

In this paper, Γ is the initial hypothesis/axiom set. Whether a member of Γ is called an axiom or an input hypothesis does not affect the proof definition: it is an admissible initial line. The same framework proves that $\text{Con}_F(\Gamma)$ is the least inference-closed set containing Γ and that rule-driven ATP runs remain inside it [13]. Those results are exactly what the Hilbert probe will reuse. The geometry does not redefine proof.

3 The Hilbert facts actually used

The classical Hilbert curve is a continuous surjection

$$H : [0, 1] \twoheadrightarrow [0, 1]^2.$$

Thus, for every point q in the unit square, there exists at least one parameter value t such that $H(t) = q$ [1]. This is the exact-coverage fact behind the search-and-rescue analogy: an idealized target modeled as a point is attained, not merely approached.

For computation, however, Data-ATP uses finite Hilbert traversals. Arbitrary-dimensional Hilbert-type encodings and traversals are well developed [2, 3]. At resolution p in dimension k , write

$$H_{k,p} : \{0, \dots, 2^{pk} - 1\} \longrightarrow \{0, \dots, 2^p - 1\}^k$$

for a finite Hilbert ordering of the grid. We require only two finite properties:

1. **coverage:** every grid cell occurs exactly once; and
2. **face adjacency:** consecutive cells differ by one unit along exactly one coordinate axis.

Equivalently, for consecutive grid points h_m and h_{m+1} ,

$$h_{m+1} - h_m = \pm e_j$$

for some coordinate basis vector e_j .

No theorem below requires a claim about measure preservation, dimension preservation, or differentiability.

The limiting Hilbert curve cannot be represented exactly by a finite polygonal drawing; figures and executable schedules necessarily use finite approximants. The proof-theoretic argument depends on surjective/exhaustive coverage and finite axis-adjacent traversal, not on the more exotic analytic properties of the limit.

4 From Hilbert coordinates to inference actions

Let $C = (\theta_0, \dots, \theta_{M-1})$ be a finite ordered list of currently verified formulas. For a rule $r_i \in z$ of arity k_i , the explicit premise-choice component is

$$\mathcal{A}_i(C) = \{i\} \times C^{k_i}.$$

Different rules are alternatives, so the natural heterogeneous action space is the tagged coproduct

$$\mathcal{A}_F(C) = \bigsqcup_i \mathcal{A}_i(C). \quad (1)$$

A production prover may require extra independently varying coordinates for substitutions, unifiers, type instantiations, or side conditions. If so, an auxiliary space $U_i(C)$ is appended and the component becomes $\{i\} \times C^{k_i} \times U_i(C)$. Thus rule arity is the base premise dimension, not automatically the entire production dimension.

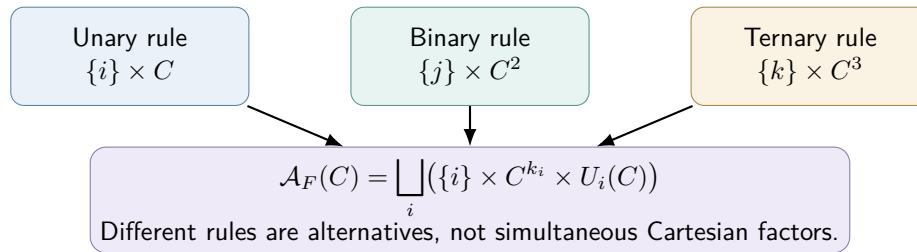


Figure 1: Rule-specific components. Arity supplies the base premise dimension; a production prover may require auxiliary coordinates $U_i(C)$ for substitutions, unification, or side-condition choices.

If $M = 0$, no positive-arity rule has a premise tuple to enumerate. Assume $M \geq 1$ for the grid construction. Choose $p = \lceil \log_2 M \rceil$ and $N = 2^p$, with $N = 1$ when $M = 1$. A finite Hilbert traversal of the k_i -dimensional grid supplies an order on candidate premise ranks. Coordinates containing an index $\geq M$ are padding and are skipped.

Proposition 4.1 (Finite rule-component coverage). *For fixed finite C and a k -ary rule, a complete finite Hilbert traversal of the padded N^k grid, skipping only padding cells, visits every ordered tuple in C^k exactly once.*

Proof. The finite Hilbert ordering is a bijection onto the full grid $\{0, \dots, N-1\}^k$. The valid subgrid $\{0, \dots, M-1\}^k$ is therefore visited exactly once at every point. The rank map $j \mapsto \theta_j$ is injective on valid indices, so every ordered premise tuple in C^k is visited exactly once. $\square$

5 Candidate points, theorem points, and one-coordinate motion

A Hilbert grid point by itself is not yet a theorem. It is an address for a candidate rule application.

Definition 5.1 (Candidate point). At verified state C , a candidate point is a tagged address

$$a = (i, \alpha_1, \dots, \alpha_{k_i}) \in \mathcal{A}_F(C).$$

Its evaluation is defined exactly when the corresponding inference rule is defined on the selected premises.

Definition 5.2 (Theorem records and theorem points). A trusted theorem record is of exactly one of the following two forms.

1. An *initial theorem record* has value $\beta \in \Gamma$. It requires no generated inference address.
2. A candidate point $a = (i, \alpha_1, \dots, \alpha_{k_i})$ is a *generated theorem point* with value β when

$$r_i(\alpha_1, \dots, \alpha_{k_i}) = \beta$$

and the designated verification layer accepts the rule instance or its certificate.

Thus every noninitial theorem record is backed by a verified direct-consequence event.

This definition makes the blog post’s geometric intuition precise. A face-adjacent Hilbert move changes one premise-rank coordinate while the remaining premise slots stay fixed. A turn changes which premise slot is varied. The path therefore generates a stream of locally related *candidate inference applications*. The logical event occurs at a successful vertex: a defined, verified rule instance produces a direct consequence.

Lemma 5.1 (Axis-parallel candidate mutation). *Inside a fixed k -ary rule component, consecutive valid finite Hilbert cells differ in at most one premise-rank coordinate before padding is removed. For consecutive face-adjacent valid cells, exactly one premise slot changes by one rank unit.*

Proof. Face adjacency gives $h_{m+1} - h_m = \pm e_j$. Decoding grid coordinate j by the ordered theorem list changes only the corresponding premise slot. If padding cells intervene, the next *valid* decoded point need not be face-adjacent; the statement is therefore local to consecutive valid Hilbert cells without skipped padding between them. $\square$

Proposition 5.2 (Theorem-point soundness). *Assume $C \subseteq \text{Con}_F(\Gamma)$. If a generated theorem point at state C has value β , then $\beta \in \text{Con}_F(\Gamma)$.*

Proof. By definition, $\beta = r_i(\alpha_1, \dots, \alpha_{k_i})$ for premises $\alpha_j \in C$. Each premise is derivable from Γ . Appending the valid rule application to proofs of the finitely many premises gives a finite proof of β from Γ . $\square$

The axis-parallel property is therefore not a substitute for inference validity. Its role is structural: it makes the probe mutate one encoded argument at a time while traversing the candidate-action space. The rule semantics and verifier determine which visited points become proof lines.

6 The Hilbert trace is a proof-generating run

Let $C_0 = \Gamma$. During epoch e , freeze C_e , traverse every finite rule component, stage all accepted new theorem points, and commit them only after the epoch completes. Define

$$D_F(C) = C \cup \{\beta \in y : \beta \text{ is a direct consequence of } C\}. \quad (2)$$

Theorem 6.1 (Hilbert-trace soundness). *Suppose every newly committed formula is the value of a verified generated theorem point whose premises were already in the current verified state. Then for every epoch e ,*

$$C_e \subseteq \text{Con}_F(\Gamma).$$

Moreover, recording the parent premises and rule label for each new formula yields a finite proof predecessor graph from which a formula-sequence proof can be reconstructed.

Proof. Induct on epochs. The base state is $C_0 = \Gamma \subseteq \text{Con}_F(\Gamma)$. If the claim holds for C_e , Proposition 5.2 places every newly accepted conclusion in $\text{Con}_F(\Gamma)$, so the enlarged state remains inside the consequence set. Each accepted node has finitely many parents from an earlier verified state; recursively expanding those parents terminates at members of Γ and yields a finite derivation DAG. A topological ordering of that finite DAG is a proof sequence in the formal definition above. $\square$

This is Data-ATP’s basic soundness argument. Search geometry can change which candidate is tried next, but it cannot make an invalid point a theorem point if acceptance remains rule-driven and verifier-controlled. This reproduces the rule-driven soundness principle already established in *Depths* [13].

Theorem 6.2 (Complete epoch equals direct-consequence closure). *Assume at epoch e that C_e is finite and the modeled rule set z is finite. Assume every rule component $C_e^{k_i}$ is completely traversed and the verifier accepts exactly the values of defined rule instances. Then*

$$C_{e+1} = D_F(C_e).$$

Proof. Every committed new formula is a legal direct consequence, giving $C_{e+1} \subseteq D_F(C_e)$. Conversely, every direct consequence of C_e has a witnessing finite rule instance. By the complete component-coverage assumption, the corresponding action address is visited. The rule is evaluated and the legal conclusion is accepted, so $D_F(C_e) \subseteq C_{e+1}$. $\square$

Theorem 6.3 (Exhaustive finite-proof discovery). *Under the hypotheses of Theorem 6.2 at every epoch, and assuming every required rule/action component is repeatedly completed fairly,*

$$\bigcup_{e \geq 0} C_e = \text{Con}_F(\Gamma).$$

Consequently, every theorem with a finite derivation in the modeled calculus appears after finitely many epochs.

Proof. Let $W = \bigcup_{e \geq 0} C_e$. Hilbert-trace soundness gives $W \subseteq \text{Con}_F(\Gamma)$. Also $\Gamma \subseteq W$. If β is a direct consequence of W , only finitely many premises witness the rule application. Those premises all occur together in some finite stage C_e because the states are monotone. Theorem 6.2 then puts β in $C_{e+1} \subseteq W$.

Thus W is inference-closed and contains Γ . Since $\text{Con}_F(\Gamma)$ is the least such set, $\text{Con}_F(\Gamma) \subseteq W$ [13]. Therefore equality holds. $\square$

Theorem 6.3 is the computational counterpart of the surjection intuition. The limiting curve’s onto property says that no point of its codomain is excluded. The executable ATP statement is stronger in the way that matters operationally: every *discrete inference address needed by a finite derivation* is reached after a finite amount of completed finite enumeration, once its premises have entered the verified state.

Corollary 6.4 (Proof-point correspondence under exhaustive Hilbert trace). *Under the hypotheses of Theorem 6.3, for every $\varphi \in y$,*

$$\Gamma \vdash_F \varphi \iff (\exists e < \infty) \varphi \in C_e.$$

If $\varphi \notin \Gamma$, its first appearance is a generated theorem point, and the recorded predecessor data reconstruct a finite proof of φ in the sense of Section 2.

Proof. The forward implication is Theorem 6.3. The reverse implication is Hilbert-trace soundness. For a noninitial formula, Theorem 6.1 also supplies the finite predecessor graph and hence a formula-sequence proof. $\square$

Corollary 6.4 is the precise formal answer to the phrase “the probe found the theorem point”: an accepted noninitial hit is not merely near a proof or suggestive of one; it comes with a reconstructible finite derivation.

The connection to the author’s proof-horizon work is immediate. A theorem has a finite birth epoch when its finite derivation becomes available; the first such epoch is a Hilbert-ordered version of a proof horizon [16]. The Hilbert order does not change theoremhood. It changes the chronology within and, under adaptive policies, among the finite horizons.

7 Budgeted Data-ATP: proof or budget exhaustion

Real runs do not receive infinitely many epochs. Fix an expansion budget $B \in \mathbb{N}$. One committed expansion is one attempted decoded inference action under the declared accounting rule.

Theorem 7.1 (Finite-budget contract). *Assume the implementation performs at most one finite amount of work per counted expansion and stops when either a verified target proof is accepted or the counter reaches B . Then the run terminates with exactly one of the following declared outcomes:*

1. **PROOF FOUND:** *a verifier-accepted proof certificate for the target; or*
2. **BUDGET EXHAUSTED:** *no accepted target proof was found within the declared budget.*

If the run returns PROOF FOUND, the result is sound under Theorem 6.1. The second outcome does not imply that the target is unprovable.

Proof. The expansion counter is bounded by B , so the declared procedure halts no later than the exhaustion event unless it halts earlier on success. Soundness of success follows from the rule-driven

theorem-point invariant. Exhaustion records a finite negative search result only; it makes no statement about actions not reached within the budget. $\square$

Thus three properties must remain separate:

soundness $\neq$ bounded termination $\neq$ exhaustive positive completeness.

Soundness comes from accepted direct consequences and verification. The budget gives bounded termination. Repeated fair complete coverage gives eventual discovery of every finite proof in the modeled calculus. None of these claims turns Data-ATP into a total decision procedure for an arbitrary sufficiently expressive undecidable theory [10, 11, 5].

8 Addresses are not yet a proof metric

The public post uses rational coordinates because Tenneson's earlier work gives an explicit surjection from the positive integers onto the rationals, together with a least-preimage section [15]. The companion proof-horizon manuscript supplies the separate chronology viewpoint used here for first appearance and finite search horizons [16]. This supplies a convenient bridge among countable addresses:

Hilbert index $\rightarrow$ grid coordinate $\rightarrow$ rational address $\rightarrow$ natural-number address $\rightarrow$ formal syntax.

For the present theory, let

$$r : \mathbb{N}_+ \twoheadrightarrow \mathbb{Q}$$

be a computable surjection and choose the canonical section

$$s(q) = \min\{n \in \mathbb{N}_+ : r(n) = q\}.$$

A normalization can then map selected rational addresses into $X = \mathbb{Q} \cap [0, 1]$ and thence into a formula enumeration.

This is useful for deterministic provenance, reproducible addressing, and cross-document compatibility. But an address system is not automatically a scientifically meaningful metric. Two formulas with adjacent line numbers can be proof-theoretically unrelated. The current Hilbert prototype therefore distinguishes

global address/provenance $\neq$ learned or engineered proof geometry.

That distinction is one of the most important corrections to an overly literal reading of the public post.

9 The real scientific risk: proof-theoretic locality

Space-filling curves are attractive because their finite orderings preserve locality better than arbitrary permutations under various metrics, though the exact locality guarantees and distortions depend on the curve and dimension [4, 3]. In theorem proving, however, Euclidean locality is useful only if the coordinate system encodes something correlated with proof relevance.

Let $L_i(C) \subseteq \mathcal{A}_i(C)$ be the legal actions of component i . Define a feature map

$$\Phi_i : L_i(C) \longrightarrow \mathbb{R}^{d_i}, \quad \Phi_i(a) = (f_1(a), \dots, f_{d_i}(a)). \quad (3)$$

Candidate features can include formula size, symbol counts, subterm signatures, type or sort information, unification statistics, proof depth, dependency overlap, prior co-occurrence, model scores, proof-DAG embeddings, or learned representations from verified proof traces.

The scheduler then operates on the induced finite geometry rather than on raw WFF line numbers. This produces a hierarchy of questions:

$$\text{legality} \rightarrow \text{representation} \rightarrow \text{geometry} \rightarrow \text{schedule} \rightarrow \text{verification}.$$

Research questions: representation and locality

Locality question. Do there exist fixed formal libraries and benchmark families for which a stable computable representation Φ makes nearby candidate actions sufficiently correlated with proof relevance that locality-preserving traversal improves verified theorem yield or time-to-target under fixed finite budgets relative to lexicographic and uniformly random schedules?

Representation question. In such tasks, is the performance effect of choosing a proof-relevant representation typically larger than the effect of choosing among reasonable locality-preserving traversals while holding the representation fixed?

The second question is deliberately unfriendly to a curve-centric story. If representation dominates, the major research problem is discovering the right proof geometry, not choosing the most elegant curve.

10 Resolution as a resource-allocation variable

At resolution p , a d -dimensional power-of-two grid has 2^{pd} cells. Increasing resolution by one bit multiplies the number of cells by

$$2^d.$$

Thus a binary component quadruples, a ternary component octuples, and higher-dimensional refinement becomes expensive very quickly.

Resolution should therefore be allocated, not blindly maximized. A controller can ask which components or subregions produced accepted steps, closed goals, reusable lemmas, repeated illegality, or redundant exploration. A simple adaptive policy may keep local Hilbert traversal deterministic while changing only the budget and resolution assigned to regions.

Let region R have observed statistics x_R , such as verifier acceptance rate, goal-reduction rate, novelty, lemma reuse, and cost. An allocation policy can be written abstractly as

$$B_{e+1}(R) = g(B_e(R), x_R),$$

subject to a global budget constraint

$$\sum_R B_{e+1}(R) \leq B_{\text{total}}.$$

This is a resource-control layer, not a modification of inference rules.

11 How fast can Hilbert search be?

The word *fast* hides several different quantities. For this project at least four should be reported separately:

1. **target rank:** how many candidate visits occur before the first verified proof of the target;
2. **transition work:** how much state must change between successive candidate evaluations;
3. **wall-clock cost:** target rank plus evaluation, verification, guidance, remapping, and logging overhead; and
4. **coverage time:** how much work is required before every member of a declared finite candidate set has been visited.

Hilbert order can be ordinary on the first metric and optimal on the second. Confusing those statements would produce a false universal speed claim.

11.1 Random sampling, random permutations, and exhaustive coverage

Let a finite feasible set contain $n \geq 1$ candidates and one distinguished target. A uniformly random permutation visits every candidate exactly once. By Theorem 12.2, its expected target rank is $(n + 1)/2$, exactly the same as any other fixed exhaustive permutation under a uniformly distributed target.

Independent uniform sampling *with replacement* behaves differently.

Proposition 11.1 (Independent random sampling). *If each trial samples independently and uniformly from n candidates with replacement, then the expected number of trials to hit a fixed target is n . The expected number of trials to visit all n candidates is*

$$nH_n = n \sum_{j=1}^n \frac{1}{j},$$

which is asymptotic to $n \log n$.

Proof. The target hit time is geometric with success probability $1/n$, hence expectation n . After j distinct candidates have been seen, the probability that the next sample is new is $(n - j)/n$, so the expected waiting time for the next new candidate is $n/(n - j)$. Summing for $j = 0, \dots, n - 1$ gives nH_n . $\square$

Thus a complete Hilbert pass and a random permutation both achieve exact coverage in exactly n visits, while naive random sampling pays for repeats. This is a coverage advantage over sampling with replacement, not a special advantage over every other no-repeat enumeration.

11.2 A best-possible local-motion theorem

Now freeze one rule component whose candidate grid is the full power-of-two cube

$$G = \{0, \dots, m-1\}^k, \quad m = 2^p,$$

with $N = m^k$ cells. Let an exhaustive ordering be a bijection

$$\pi : \{0, \dots, N-1\} \rightarrow G.$$

Define its total grid-transition length by

$$L_1(\pi) = \sum_{j=0}^{N-2} \|\pi(j+1) - \pi(j)\|_1$$

and its total changed-slot count by

$$L_H(\pi) = \sum_{j=0}^{N-2} d_H(\pi(j+1), \pi(j)),$$

where d_H is Hamming distance between coordinate tuples.

Theorem 11.2 (Full-grid Hilbert transition optimality). *For every exhaustive ordering π of G ,*

$$L_1(\pi) \geq N-1, \quad L_H(\pi) \geq N-1.$$

A face-adjacent finite Hilbert traversal attains equality in both bounds:

$$L_1(H) = L_H(H) = N-1.$$

Hence Hilbert traversal is transition-cost optimal under either unit Manhattan motion or one-cost-per-changed-coordinate accounting. The optimum is not unique.

Proof. Consecutive entries of an exhaustive ordering are distinct. Any two distinct integer-grid points have ℓ_1 distance at least 1 and Hamming distance at least 1. There are $N-1$ transitions, so both sums are at least $N-1$. A face-adjacent Hilbert traversal changes exactly one coordinate by one unit at every transition, so each summand equals 1. $\square$

The theorem identifies a precise sense in which Hilbert search can be *best possible*. It does not say Hilbert is the unique optimum: serpentine Hamiltonian paths and other face-adjacent traversals can attain the same lower bound. It also applies to the full grid. When padding or legality filters cause cells to be skipped, consecutive *valid* candidates need not remain adjacent and the equality requires separate analysis.

Proposition 11.3 (Expected premise-slot churn under a random permutation). *Let Π be a uniformly random permutation of G . For any adjacent pair in the random ordering,*

$$\mathbb{E}[d_H(\Pi(j), \Pi(j+1))] = \frac{k(1 - 1/m)}{1 - m^{-k}}.$$

For fixed k this approaches k as $m \rightarrow \infty$, whereas a face-adjacent Hilbert traversal has changed-slot

count exactly 1 at every step.

Proof. An adjacent ordered pair in a uniformly random permutation is uniformly distributed over ordered distinct pairs (X, Y) from G . Before conditioning on $X \neq Y$, a given coordinate differs with probability $1 - 1/m$. Coordinate difference already implies $X \neq Y$, so

$$\mathbb{P}(X_i \neq Y_i \mid X \neq Y) = \frac{1 - 1/m}{1 - m^{-k}}.$$

Summing the k coordinate indicators gives the formula. □

11.3 When transition optimality becomes runtime optimality – and when it does not

Suppose an implementation can reuse computation across neighboring candidate actions and its total cost over a full component has the form

$$T(\pi) = T_0 + Nc_{\text{eval}} + c_{\text{chg}}L_H(\pi), \quad c_{\text{chg}} \geq 0, \quad (4)$$

where c_{eval} is unavoidable per-candidate work and c_{chg} is the additional cost of invalidating or recomputing one changed premise slot.

Corollary 11.4 (Conditional runtime optimality). *Under the exact cost model (4) on a full power-of-two rule grid, any face-adjacent Hilbert traversal minimizes $T(\pi)$ among exhaustive orderings.*

Proof. All terms except $L_H(\pi)$ are ordering-independent, and Theorem 11.2 minimizes L_H . □

This conditional result gives the implementation question a concrete target. If unification state, side-condition checks, parsed premises, feature calculations, or other intermediate data can be reused when only one premise slot changes, local motion can save real work. If every candidate is recomputed from scratch so that $c_{\text{chg}} = 0$, all exhaustive orders tie under this model and the Hilbert adjacency theorem produces no runtime advantage. Likewise, if verifier cost dominates everything else, transition savings can be negligible in wall-clock time.

The remaining source of possible time-to-proof improvement is *information*: a useful representation, a nonuniform target distribution, learned ranking, or privileged guidance may move productive regions earlier. The next sections separate those effects from the local transition theorem.

12 Constrained hidden-target search

The password-like analysis is useful precisely because it can expose when the Hilbert intuition fails. It should be treated as an abstract finite-search model rather than as an operational credential attack.

Let P be a finite candidate set and let hard conditions $C_1, \dots, C_m$ define the feasible subset

$$P^* = \{p \in P : C_1(p) = \dots = C_m(p) = 1\}. \quad (5)$$

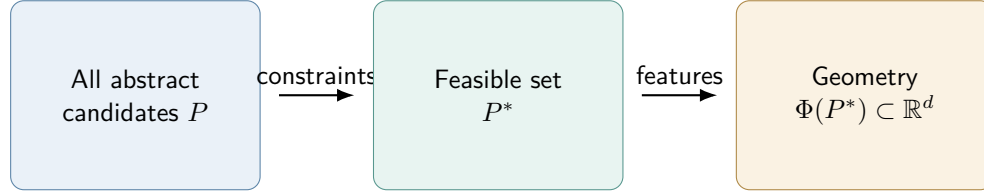
A scheduler is a permutation

$$\pi : \{1, \dots, |P^*|\} \rightarrow P^*.$$

A feature map

$$\Phi : P^* \rightarrow \mathbb{R}^d$$

can place feasible candidates into a geometry before a Hilbert or other locality-preserving order is imposed.



Only after feasibility and representation are fixed does a traversal policy choose an order.

Figure 2: The constrained hidden-target decomposition: feasibility, representation, and scheduling are distinct operations.

A “password-like” example can be understood as a finite family of abstract token strings subject to known conditions such as a fixed length or membership constraints. The mathematical point is not how to attack an account. It is that hard conditions remove impossible candidates, while richer quantitative features may distinguish neighborhoods among the survivors.

Proposition 12.1 (Pruning is scheduler-independent). *The reduction from P to P^* in (5) occurs before a visitation order is chosen. Consequently, the cardinality reduction $|P| \rightarrow |P^*|$ cannot be credited to Hilbert order or to any other scheduler that merely permutes P^* .*

Proof. Membership in P^* is defined solely by the predicates C_j . A scheduler is chosen on the already defined set P^* . Therefore the feasible-set cardinality is independent of the subsequent permutation. $\square$

Theorem 12.2 (Uniform-target permutation invariance). *Suppose the hidden target T is uniformly distributed on a nonempty finite feasible set P^* of size n , and a deterministic scheduler visits each candidate exactly once. For every scheduler permutation π ,*

$$\mathbb{E}[\text{rank}_\pi(T)] = \frac{n+1}{2}.$$

Under a budget of $b \leq n$ visits, the success probability is b/n for every such scheduler.

Proof. Uniformity assigns probability $1/n$ to the candidate at every rank j . Hence

$$\mathbb{E}[\text{rank}_\pi(T)] = \frac{1}{n} \sum_{j=1}^n j = \frac{n+1}{2}.$$

The first b positions contain exactly b of the n equally likely candidates, so success probability is b/n . $\square$

The theorem is the required negative control. Hilbert order cannot create information. A plausible advantage requires nonuniform target probability correlated with the chosen geometry, adaptive observations, or some other source of structure.

Research question: constraint-induced locality

For some finite hidden-target families, do known target conditions induce quantitative features whose neighborhoods correlate with target probability strongly enough that a coarse-to-fine Hilbert schedule outperforms an equally constraint-aware but locality-oblivious baseline under fixed finite budgets?

This question is a clean transfer back to ATP. Rule side conditions correspond to hard feasibility. Proof descriptors correspond to Φ . The Hilbert cursor corresponds only to scheduling. The decomposition prevents an improvement due to pruning from being incorrectly reported as an improvement due to geometry.

13 Search hypotheses are not axioms

Suppose a target theorem φ is known to belong to a mechanically checkable subclass Y – for example, a syntactic class, a type class, or a family with specified symbols. This information may filter or prioritize terminal candidates. It must not be silently inserted into the logical hypothesis set Γ .

Axiom versus search metadata

An axiom changes what is derivable. Search metadata changes only which already legal possibilities receive attention. Conflating the two would invalidate the experimental comparison and can invalidate soundness if the metadata is smuggled into the proof assumptions.

This distinction is especially useful in the hidden-target analogy. Knowing that a target has a property is not the same thing as proving a theorem from an additional axiom asserting that property.

14 A breadth-preserving Hilbert hybrid

Hilbert order need not replace breadth-oriented search. Let L_m denote a fixed derivational-depth layer or another breadth-controlled frontier. Within each L_m , legal candidate actions can be embedded and visited in Hilbert order. The outer policy preserves breadth; the inner policy changes only horizontal visitation among comparable actions.

This design gives an unusually clean matched experiment:

- same theorem library and target;
- same breadth layers;
- same legal candidate generator;
- same verifier;
- same expansion, time, and memory budgets;
- same learned scores, if any;
- only the intra-layer order differs.

A breadth-preserving Hilbert condition can then be compared to ordinary breadth order without confounding proof depth with geometry.

15 The restricted-advice Professor: the paper’s single conjecture

Lay version

Someone knows a route to the answer but is not allowed to tell us the route. They may only give a tightly restricted hint such as “warmer” or “colder.” How much faster can an exhaustive proof search become before the hints amount to giving away the proof?

Fix a target theorem φ and one valid witness proof π^* known to a privileged Professor but not supplied to Data-ATP. At query t , the prover exposes a declared search summary s_t and receives only

$$q_t = Q(s_t, \pi^*) \in \mathcal{Q}_t.$$

The semantics of Q are fixed before the experiment: for example, a one-bit warmer/colder response or a small quantized progress score. If at most q queries are permitted and $|\mathcal{Q}_t| \leq 2^b$, the transcript has a declared communication budget $B \leq qb$ bits. This resembles advice-complexity models, where privileged input knowledge is supplied through a bounded advice channel [9], but here the resource guides proof search.

Hypotheses needed for a useful secret-keeping Professor.

1. **Witness access.** A finite valid proof π^* exists and is available to the Professor.
2. **Bounded semantic channel.** The scoring rule, quantization, query count, and output alphabet are predeclared. The Professor cannot output raw proof lines, full premise tuples, rule-instance identifiers, or the next correct inference. This is bounded leakage, not zero knowledge.
3. **Informativeness.** The permitted score is statistically related to progress toward at least one successful proof route. An independent or adversarial signal should not help systematically.
4. **Scheduler-only authority.** Advice may change order, region, resolution, or budget allocation, but never theorem acceptance. The verifier remains authoritative.
5. **Fair fallback.** If exhaustive positive completeness is claimed, advice cannot starve required finite regions forever.
6. **Full accounting and clean separation.** Professor compute, communication, and bookkeeping count toward cost, and the hidden proof or equivalent labels do not leak into the unguided baseline or its training data.

Conjecture 15.1 (Restricted-Advice Professor Conjecture). *There exist effective finitary proof-search families, target distributions, and predeclared bounded advice channels for which a Professor possessing a hidden valid proof reduces the expected total cost-to-proof of a verifier-separated Hilbert prover relative to the matched unguided Hilbert prover, even after advice overhead is charged. The channel does not directly transmit proof steps, and fair unbounded continuation still reaches every finite proof.*

The conjecture need not apply when no witness proof is available, the score is useless or misleading, guidance costs more than it saves, the channel effectively reveals the next proof step, the score does not align with the search representation, or fairness is abandoned. A Professor tied to one witness can also steer away from a different, cheaper proof.

The experiment asks how many advice bits and how much advice computation are needed to save how many expansions or seconds.

A natural ladder is 0 bits, one-bit warmer/colder advice, a few-bit score, a richer ranking, and an unrestricted proof-revealing oracle. The last condition is a control, not an interesting search algorithm.

16 Creativity as verified structural novelty

A separate Data-ATP research thread treats creativity as an experimentally controlled property of search rather than as a synonym for randomness [17]. Fix a formal system F , hypotheses Γ , target φ , verifier V , information condition K , training exposure p , and resource protocol T . Let a policy configuration be

$$u = (\tau, w, b, n, r, \lambda, \ell),$$

where the coordinates may control temperature, candidate width, breadth-depth allocation, novelty pressure, restarts, intermediate-lemma construction, and proof-length preference.

Let S_u be the event that policy u finds a verifier-accepted proof. Let $\nu(P_u) \in [0, 1]$ be a predeclared structural-novelty score after proof normalization, and set novelty to zero on failure. Define the realized creative return

$$Z_u = \mathbf{1}_{S_u} \nu(P_u)$$

and expected creative yield

$$\mathcal{C}(u) = \mathbb{E}[Z_u].$$

Let

$$D(u) = \mathbb{P}(S_u), \quad N(u) = \mathbb{E}[\nu(P_u) \mid S_u]$$

with $N(u) = 0$ if $D(u) = 0$.

Theorem 16.1 (Creative-yield factorization). *For every policy u ,*

$$\mathcal{C}(u) = D(u)N(u).$$

Proof. Conditioning on S_u gives

$$\mathbb{E}[\mathbf{1}_{S_u} \nu(P_u)] = \mathbb{P}(S_u) \mathbb{E}[\nu(P_u) \mid S_u].$$

The zero-success case is zero by definition. □

This prevents a common interpretive error. A policy that produces bizarre, highly novel proof fragments but almost never proves anything need not have high creative yield. Likewise, a policy that reliably reconstructs the same conventional proof can have high success and low novelty.

Hilbert geometry is therefore not “creativity” by definition. It is one possible mechanism for redistributing attention. Its value must be measured by outcomes such as success, cost, structural novelty, robustness, and transfer.

17 Multi-agent search and changing terrain

Once verified lemmas are shared, multiple search agents no longer traverse independent static spaces. Let agent a at epoch e have policy $\pi_{a,e}$, budget $B_{a,e}$, and possibly a region or rule assignment $R_{a,e}$. If agent a contributes a verified lemma λ , the shared set becomes

$$C_{e+1} = C_e \cup \{\lambda\} \cup \dots,$$

which can change every other agent’s action components $\mathcal{A}_i(C_{e+1})$ and feature geometry Φ_i .

Agents can differ in resolution, starting region, exploration pressure, lemma seeking, learned ranking, and willingness to revisit old regions. A scheduler may allocate resources according to marginal verified value rather than attempting to make agents uniformly different.

Research hypothesis: shared-lemma population gain

There may exist theorem families and fixed total budgets for which a heterogeneous population of verifier-separated agents sharing accepted intermediate lemmas achieves a higher probability of target proof, or a lower expected cost to proof, than a single matched agent consuming the same total budget.

This hypothesis must be tested against a strong single-agent baseline. Merely splitting a budget across agents does not create information, and synchronization overhead can erase any gain.

18 Data-ATP Phase 0.0.1 as a baseline, not a result

The public post describes the present experimental bridge conservatively. Predator 8.004 supplies the machine-guided proof search, while Data-ATP supplies persistent experimental infrastructure: committed-expansion accounting, transaction logging, checkpoints, deterministic replay, crash recovery, and “breadcrumbs” that allow a long search to be interpreted after interruption [12].

At the snapshot reported in the public post, the target was the Metamath theorem `sgrpc1` under a fixed budget of 80,000 expansions, with the target proof and downstream theorems excluded from training. The bridge intentionally used one Predator agent. The reported run had passed 13,000 committed expansions and remained dynamically active.

Those facts are valuable because they define an operational baseline. They are *not* evidence that Hilbert geometry works: the long-running conservative condition was not a matched Hilbert-versus-non-Hilbert experiment. The scientific role of the baseline is to establish logging, restart fidelity, budget semantics, and a reference condition that later geometry must beat.

19 Experimental protocols

19.1 GEO-01: isolate the scheduler

Protocol 19.1 (GEO-01: matched geometry pilot). Freeze the formal environment, theorem library, target set, candidate generator, legality checks, verifier, training data, model weights, expansion definition, wall-clock cap, memory cap, seeds, and stopping rules. Compare only predeclared scheduler/representation conditions.

A useful initial matrix is:

1. lexicographic enumeration;
2. uniformly randomized order with recorded seeds;
3. ordinary breadth-oriented traversal;

4. the existing learned/heuristic baseline;
5. Hilbert order on a naive deterministic address such as WFF rank;
6. Hilbert order on a predeclared proof-feature representation Φ ;
7. breadth-preserving Hilbert order within fixed layers;
8. optionally, a second space-filling baseline such as Morton/Z-order to distinguish “Hilbert” from “locality-preserving linearization” in general.

Primary endpoints should include solve rate, committed expansions to proof, wall-clock time, peak memory, verified intermediate lemmas, proof length, candidate rejection rate, repeated/redundant actions, and robust performance across repeated seeds.

The most important ablation is representation versus scheduler. If Hilbert on naive WFF rank fails while Hilbert on Φ succeeds, the result supports a geometry claim. If Hilbert and another locality-preserving order behave similarly, the mechanism may be locality rather than a Hilbert-specific property. If every geometry condition loses to learned ranking, that is also a scientifically useful result.

19.2 SPEED-01: separate target rank from transition work

Protocol 19.2 (SPEED-01: local-cost and wall-clock ablation). On full power-of-two rule grids, compare Hilbert, serpentine, lexicographic, Morton/Z-order, and seeded random permutations while holding the candidate set fixed. Record target rank, changed premise slots per transition, reusable-cache hits, verifier calls, total expansions, and wall-clock time. Then repeat with reuse deliberately disabled. Theorem 11.2 predicts that Hilbert and any other face-adjacent Hamiltonian traversal tie at the minimum changed-slot count; Corollary 11.4 predicts a runtime benefit only when changed-slot reuse has nonzero cost value.

19.3 PROF-01: restricted-advice Professor

Protocol 19.3 (PROF-01: advice-bandwidth ladder). Choose benchmark targets whose witness proofs are hidden from the base prover but available to a separate Professor process. Predeclare a semantic score Q , its quantization, query schedule, and total advice-bit budget. Compare 0-bit guidance, one-bit warmer/colder guidance, several small score alphabets, and an unrestricted proof-revealing oracle control. Charge all Professor computation and communication to the resource accounting. Preserve an unguided fairness quota whenever exhaustive fallback completeness is being tested.

Primary endpoints are expansions to verified proof, wall-clock time to verified proof, total advice bits, Professor compute time, proof length, and failure rate under the same outer resource cap. The key plot is cost-to-proof as a function of advice bandwidth. A useful Professor must beat the 0-bit baseline before the channel approaches direct proof disclosure.

19.4 DATP-CREAT-01: isolate creativity controls

The creativity manuscript proposes a staged protocol before introducing full population dynamics [17]:

- **Stage A:** validate controls and logging on inexpensive repeated targets such as `prcom`, `prcon`, `0cn`, plus a frozen sample of additional pre-target theorems;

- **Stage B:** select configurations by predeclared criteria and transfer them, without post-hoc tuning on outcomes, to `sgrpc1`;
- **Stage C:** after reproducibility is established, transfer the same mechanisms to a substantially harder held-out benchmark such as HaloProof.

The general principle is to use cheap targets to debug the experimental instrument, not to proclaim research victory.

19.5 Hidden-target negative controls

A separate finite simulator can isolate the geometry from theorem-prover complexity. Generate feasible candidate sets with controlled target distributions:

- uniform target on P^* – every permutation should match Theorem 12.2;
- clustered target probability correlated with Φ – locality may help;
- adversarial or anti-correlated geometry – locality should hurt or be neutral;
- identical constraints but different schedulers – pruning benefit must remain constant;
- identical geometry but different curves – tests curve-specific effects.

A method that cannot pass these synthetic controls should not be trusted in a more complicated ATP benchmark.

20 Reproducibility: breadcrumbs as scientific state

Long-running proof search is unusually vulnerable to ambiguous restarts. A process can terminate after hours or days, and “start again” is not the same experiment as “continue.” Data-ATP therefore treats persistent state as part of the scientific instrument.

At minimum, a reproducible run should record:

- exact code revision and dependency versions;
- theorem-library revision and training-data exclusion rule;
- target and protocol identifier;
- model/checkpoint hashes and random seeds;
- committed expansion count and the exact expansion definition;
- transaction/event log with monotone identifiers;
- frontier or reconstructible search state;
- accepted certificate/provenance records;
- scheduler position, component, resolution, and any adaptive controller state;
- wall-clock, CPU/GPU, memory, and interruption events;

- recovery action and evidence that replay reconstructs the pre-interruption state.

A hash-chained event log is attractive because it makes accidental deletion/reordering detectable, but cryptographic logging is not required for the mathematical idea. The essential property is that a later researcher can distinguish a faithful continuation from a new trial.

21 Prior-art review and originality boundary

21.1 What is established

The surrounding ingredients are established. Hilbert’s curve supplies the classical space-filling surjection [1]; computational and higher-dimensional Hilbert traversals are standard topics [2, 3]; discrete space-filling curves have studied locality properties [4]; and modern ATP systems use many heuristic and machine-learned mechanisms to choose what to explore next [5, 7]. Formal proof certificates and small verifiers are likewise standard practice in proof-producing systems, including Metamath and proof-producing SAT workflows [6, 8].

The present paper therefore does not claim invention of space-filling curves, locality-preserving order, theorem proving, proof verification, machine guidance, or finite exhaustive search.

21.2 An important close precedent found on re-review

A renewed literature search found a result that must be included in any serious novelty discussion. Rostami and Bright’s 2025 paper *Queen Domination by SAT Solving* uses a Hilbert-curve ordering of chessboard Boolean literals inside cardinality-constraint encodings. Their SAT workflow is proof-producing: solver output is checked by an independent LRAT proof checker, and the paper reports substantial runtime gains from the Hilbert literal ordering on its queen-domination instances [8].

This is closer prior art than the earlier internal versions recognized. It means the broad statement “Hilbert ordering has not been used in proof-related search” is false and must not appear in this manuscript.

The construction here is nevertheless different in object and role. Rostami and Bright use spatial Hilbert order to arrange input literals so that a conventional SAT solver receives a better encoding. Data-ATP proposes Hilbert traversal over the *inference-action space itself*: a rule tag chooses a component, premise coordinates choose antecedent formulas, optional auxiliary coordinates choose other rule-instance variables, and visiting a successful point produces a direct consequence. Repeated complete epochs are then proved to recover the deductive closure of the modeled finitary calculus.

21.3 Targeted search result after narrowing the claim

The review for this edition used combinations of “Hilbert curve,” “space-filling curve,” “Hilbert order,” “proof search,” “automated theorem proving,” “automated reasoning,” “inference rule,” “premise selection,” SAT, SMT, and proof generation. It also specifically searched for Hilbert traversal of inference-rule tuples or proof-action coordinates. The closest located item was the Rostami–Bright SAT literal-ordering paper above. No reviewed source was found that presents the same rule-specific construction in which finite Hilbert traversal directly enumerates tagged inference addresses and in which complete-epoch and repeated-closure theorems are derived from that enumeration.

A negative literature search is not proof of universal priority. Terminology varies, older work can be hard to index, and neighboring SAT/SMT literature is large. The novelty statement must therefore remain conditional on the review.

21.4 Claim-by-claim originality assessment

Claim / component	Status after review	Defensible wording
Hilbert space-filling surjection and finite traversals	Classical / established	Background only; no originality claim.
Hilbert order in proof-related SAT computation	Prior art exists: Rostami–Bright (2025)	Explicitly acknowledge; broad priority claim is unavailable.
Rule arity as base premise dimension	Natural formal construction	Useful definition, not claimed as a major standalone invention.
Tagged coproduct of heterogeneous rule-specific premise spaces	Mathematically natural; exact ATP use not located	Part of the proposed inference-address architecture.
Hilbert traversal directly over rule-premise inference addresses	No matching construction located in targeted review	Narrow candidate originality claim.
Theorem-point semantics: a successful visited action is a direct consequence	Derived here from the formal-system definition	Project-specific bridge from geometry to proof.
Complete Hilbert epoch equals direct-consequence closure	Proved under explicit finite-action assumptions	Mathematical result about this construction.
Repeated fair epochs enumerate $\text{Con}_F(\Gamma)$	Follows from least inference closure once complete epochs are established	Consequence/completeness theorem for this architecture, not a general new logic theorem.
Full-grid minimum transition cost	Elementary consequence of face adjacency	Proven optimality only for the declared local transition metric; not unique and not universal time-to-proof optimality.
Restricted-advice Professor	Related advice-complexity theory exists; ATP speed claim unproved	Single conjectural research direction with bounded communication and full cost accounting.
Password-like constrained hidden-target transfer model	Constrained search is established; exact transfer model is project-specific	Use as a controlled analogy and negative-control framework, not as a priority claim about password search.

21.5 The originality statement this paper can support

Prior-art statement. Hilbert ordering has previously appeared in proof-producing SAT computation, notably as a literal-ordering strategy in Rostami and Bright (2025). The narrower construction proposed here is different: finite Hilbert traversal is applied directly to tagged, rule-specific inference-action addresses, and successful visited points are interpreted as direct consequences so that completed epochs recover one-step closure. To the best of the targeted literature review reported here, no prior publication was located with this exact inference-address construction. This is a qualified negative-search result, not proof of priority.

That wording supports describing the manuscript as containing potentially original research while avoiding a claim that Hilbert curves themselves, Hilbert ordering in SAT, or heuristic proof search are new.

22 Falsifiers and failure modes

A credible research proposal should explain what would make it fail.

1. **Bad geometry.** If candidate neighborhoods do not correlate with proof relevance, Hilbert locality preserves the wrong thing.
2. **Dimensional explosion.** High-arity or auxiliary-coordinate components may become too expensive for useful resolution.
3. **Learned ranking dominance.** A strong learned policy may exploit the same information more directly than a geometric traversal.
4. **Curve irrelevance.** Morton order, lexicographic blocks, or cache-friendly batching may match Hilbert performance, showing that the benefit is generic locality or implementation locality rather than Hilbert structure.
5. **Dynamic invalidation.** Rapidly changing theorem sets may make frozen epochs or stale geometries wasteful unless remapping is inexpensive.
6. **Constraint-only illusion.** Improvements in hidden-target models may come entirely from pruning, not from traversal order. Proposition 12.1 is designed to detect this.
7. **No-information invariance.** Under uniform targets, any claimed average Hilbert advantage violates the negative control unless the experimental conditions differ elsewhere.
8. **No reusable transition state.** If candidate inference attempts are recomputed independently, the transition-optimality theorem may have essentially no wall-clock consequence.
9. **Advice without information.** A Professor whose bounded signal is independent of productive proof routes should not systematically beat the unguided condition.
10. **Advice leakage.** If a purportedly small guidance channel effectively identifies the next proof step, the experiment has become proof transmission rather than search guidance.
11. **Overhead.** Coordinate computation, remapping, Professor queries, logging, multi-agent synchronization, or verifier traffic may cost more than the search savings.
12. **Non-transfer.** Gains on one theorem family may disappear on held-out families, indicating overfit geometry.

A negative result on WFF-rank geometry would reject that representation, not every possible Hilbert-ordered ATP. Conversely, a positive result must survive representation and curve ablations before it should be attributed specifically to Hilbert order.

23 Research roadmap

The shortest responsible path from the current baseline to a serious claim is:

1. freeze and publish the exact expansion/budget semantics;
2. run synthetic hidden-target controls, including Theorem 12.2;
3. validate the transition-cost accounting of Section 11 on full grids, including a face-adjacent non-Hilbert control;
4. implement a minimal rule-component Hilbert scheduler with no adaptive controller;
5. compare it to lexicographic, seeded random, breadth, and existing learned/heuristic baselines;
6. separate naive address geometry from feature geometry;
7. add one alternate locality-preserving curve as an ablation;
8. only after reproducible gains, introduce adaptive resolution;
9. only after unguided geometry is characterized, run the predeclared restricted-advice Professor ladder of Section 15;
10. only after adaptive gains, introduce heterogeneous multi-agent allocation and shared lemmas;
11. preserve held-out transfer tests so that geometry is not tuned directly on the final benchmark;
12. require final theorem certificates to be accepted by the designated external verifier regardless of search policy.

This order follows the project’s own design rule: complexity should earn its way into the architecture.

24 Conclusion

The technical version of the Data-ATP blog post can now be stated without metaphor. A formal system supplies well-formed formulas and finitary partial inference rules. A finite verified state supplies the premises available now. Rule arity creates a natural premise-coordinate space, and heterogeneous rules form a tagged coproduct. A finite Hilbert traversal exhausts each discretized rule component, and its underlying grid path changes one coordinate at a time between face-adjacent cells. After padding cells are skipped, successive valid decoded candidates need not themselves remain adjacent. Each valid visited point is decoded into a candidate rule instance. When the rule is defined and verification accepts the result, the point produces a direct consequence and therefore a legitimate new proof line.

This yields the central mathematical chain:

$$\boxed{\text{Hilbert coverage} \longrightarrow \text{inference addresses} \longrightarrow \text{direct consequences} \longrightarrow \text{finite proofs.}}$$

A complete frozen epoch computes the direct-consequence closure of the current state. Repeated fair complete epochs enumerate the entire finite-proof consequence set of the modeled calculus. Under a finite budget, Data-ATP either returns a verified proof or reports budget exhaustion; failure within budget is not unprovability. Soundness comes from rule-driven verified acceptance, while exhaustive coverage supplies positive completeness.

Speed is no longer a single undifferentiated question. Hilbert order cannot create information, and the uniform-target negative control proves that no exhaustive permutation has an expected-rank advantage under complete ignorance. Yet on a full power-of-two rule grid, face-adjacent Hilbert traversal attains the absolute lower bound on total one-coordinate transition work. Under an exact incremental-reuse cost model that local property becomes runtime-optimal, though not uniquely so. If adjacent candidate evaluations share no reusable state, that advantage disappears.

The paper's one conjecture asks what happens when information is added in a controlled way. A restricted-advice Professor may know a valid proof but communicate only through a predeclared bounded semantic channel. The verifier still certifies every accepted step, all Professor cost is charged, and fair fallback preserves positive completeness. The resulting research question is quantitative: how many bits of privileged guidance are needed to save how many expansions or seconds without simply transmitting the proof?

The prior-art review also changed the manuscript in an important way. Hilbert ordering has already been used in a proof-producing SAT workflow, so the project cannot claim that Hilbert geometry has never touched proof search. The narrower inference-address construction remains distinct in the literature reviewed here. That is the claim this paper invites experts to attack, refine, reproduce, or falsify.

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